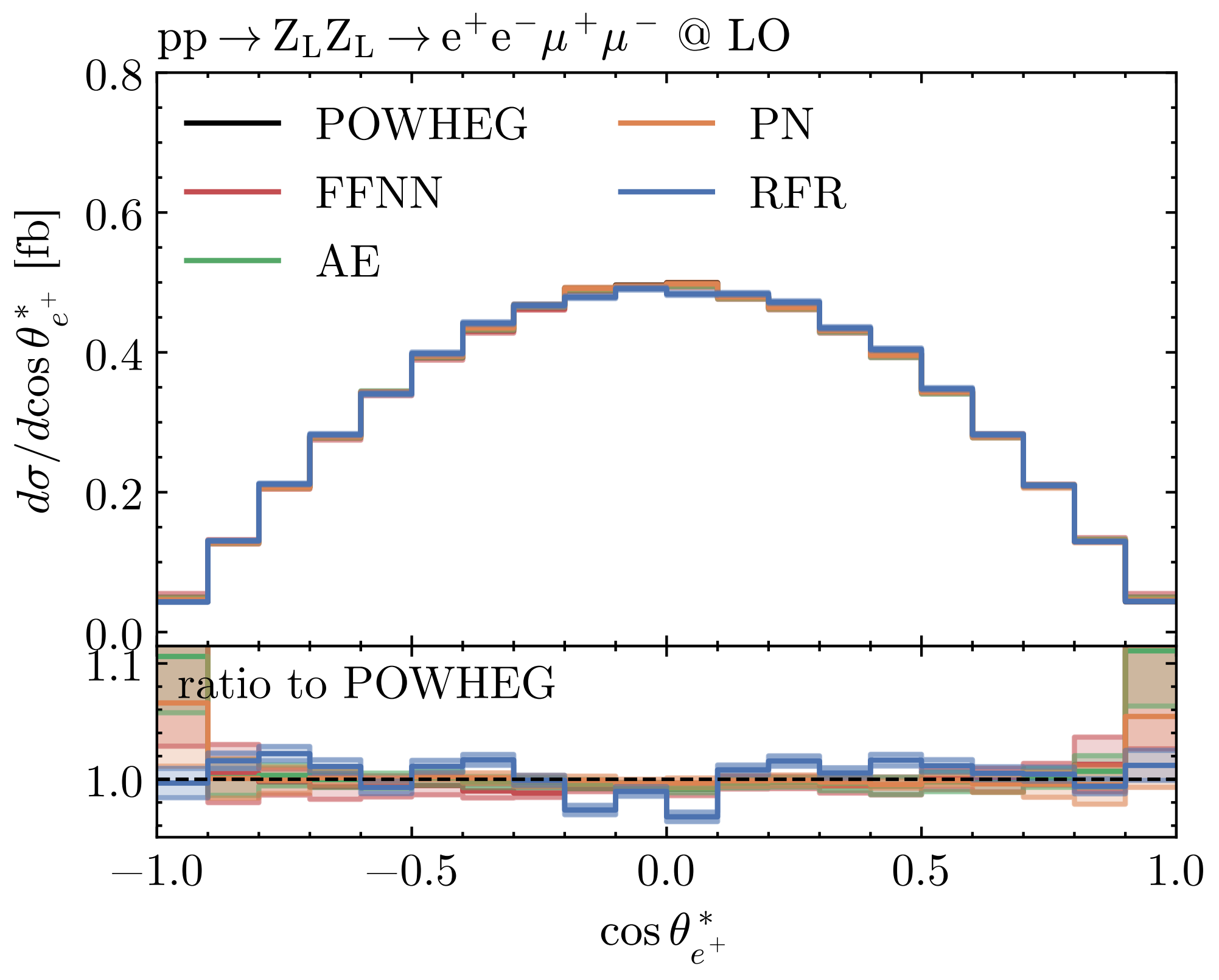
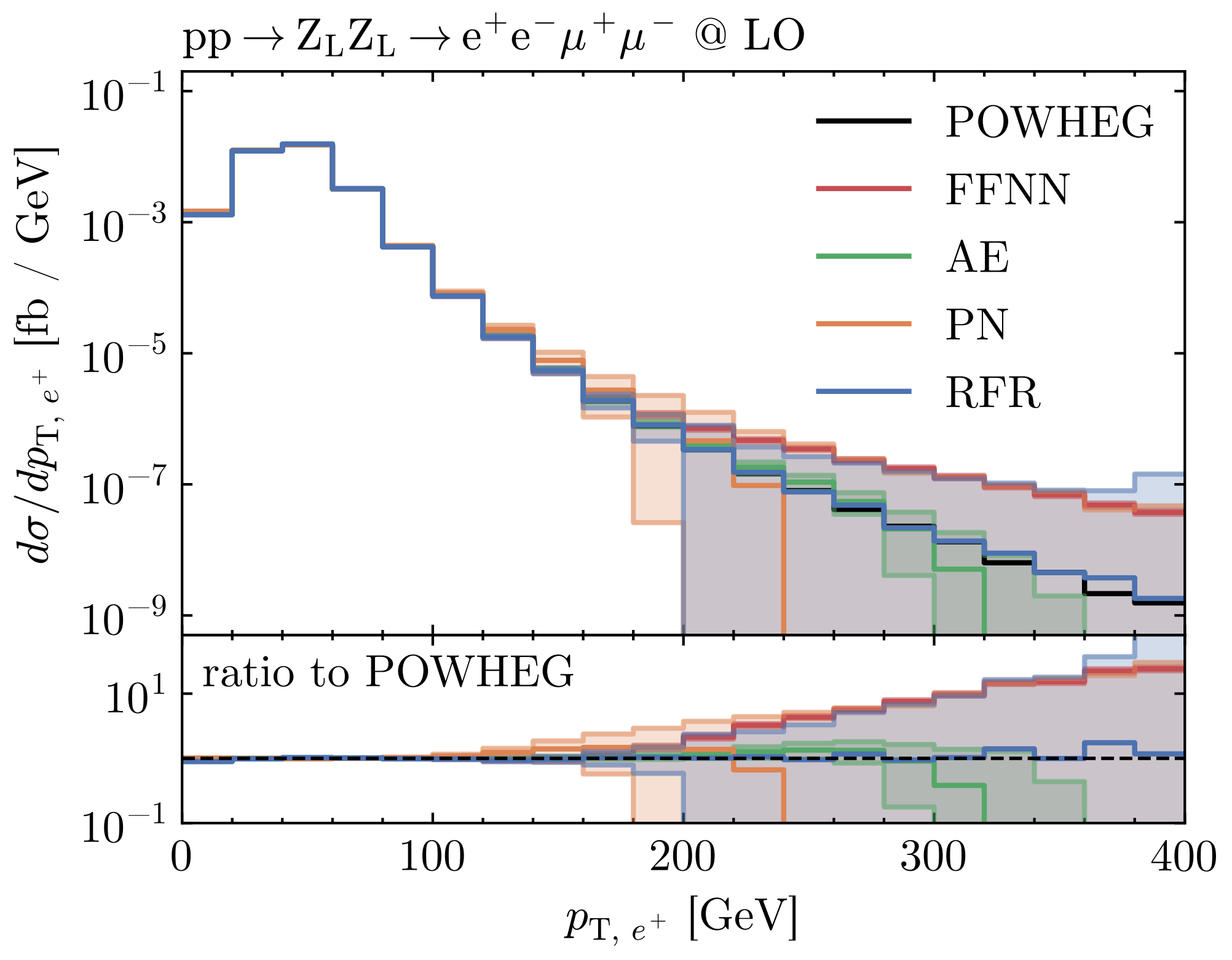
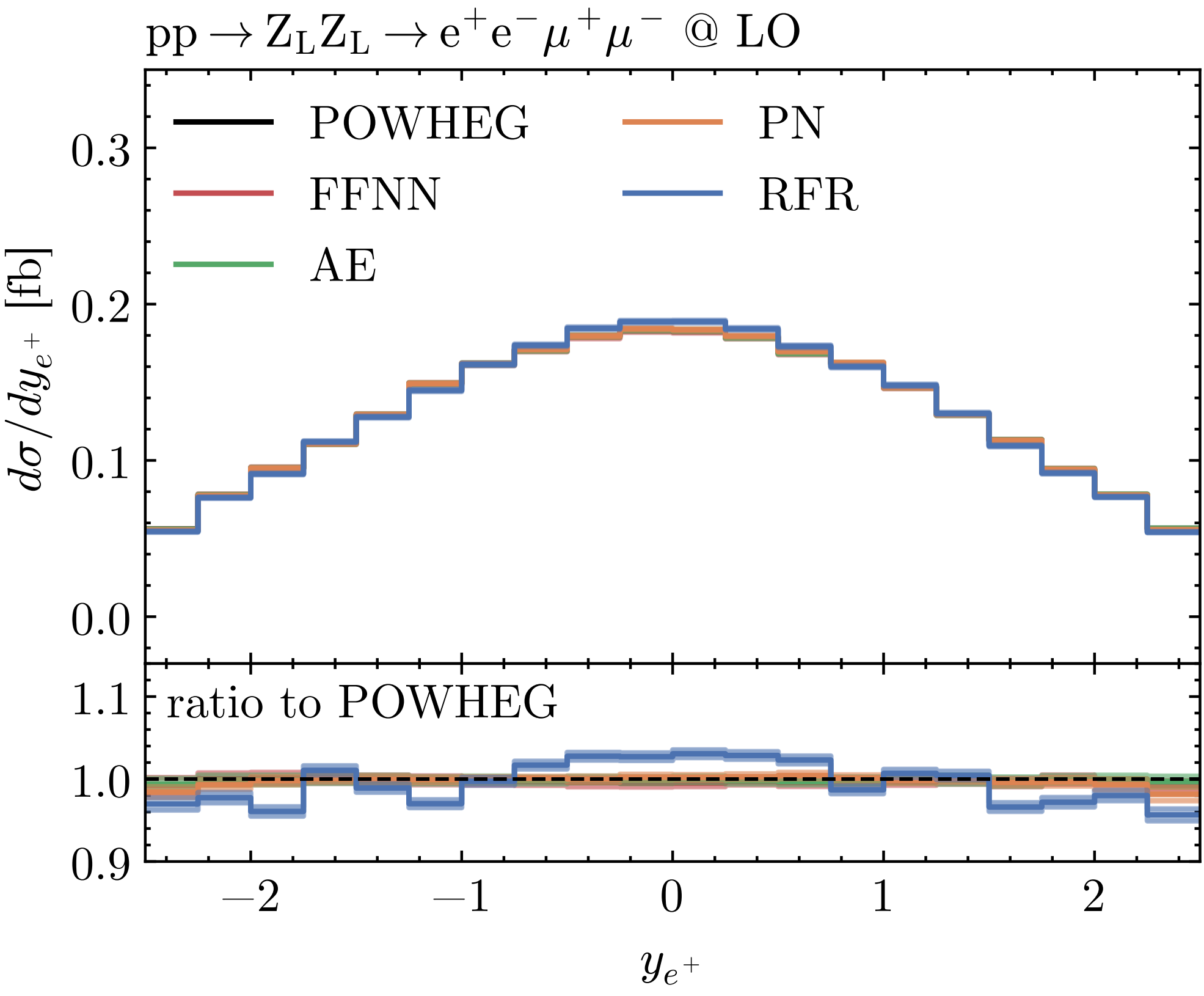
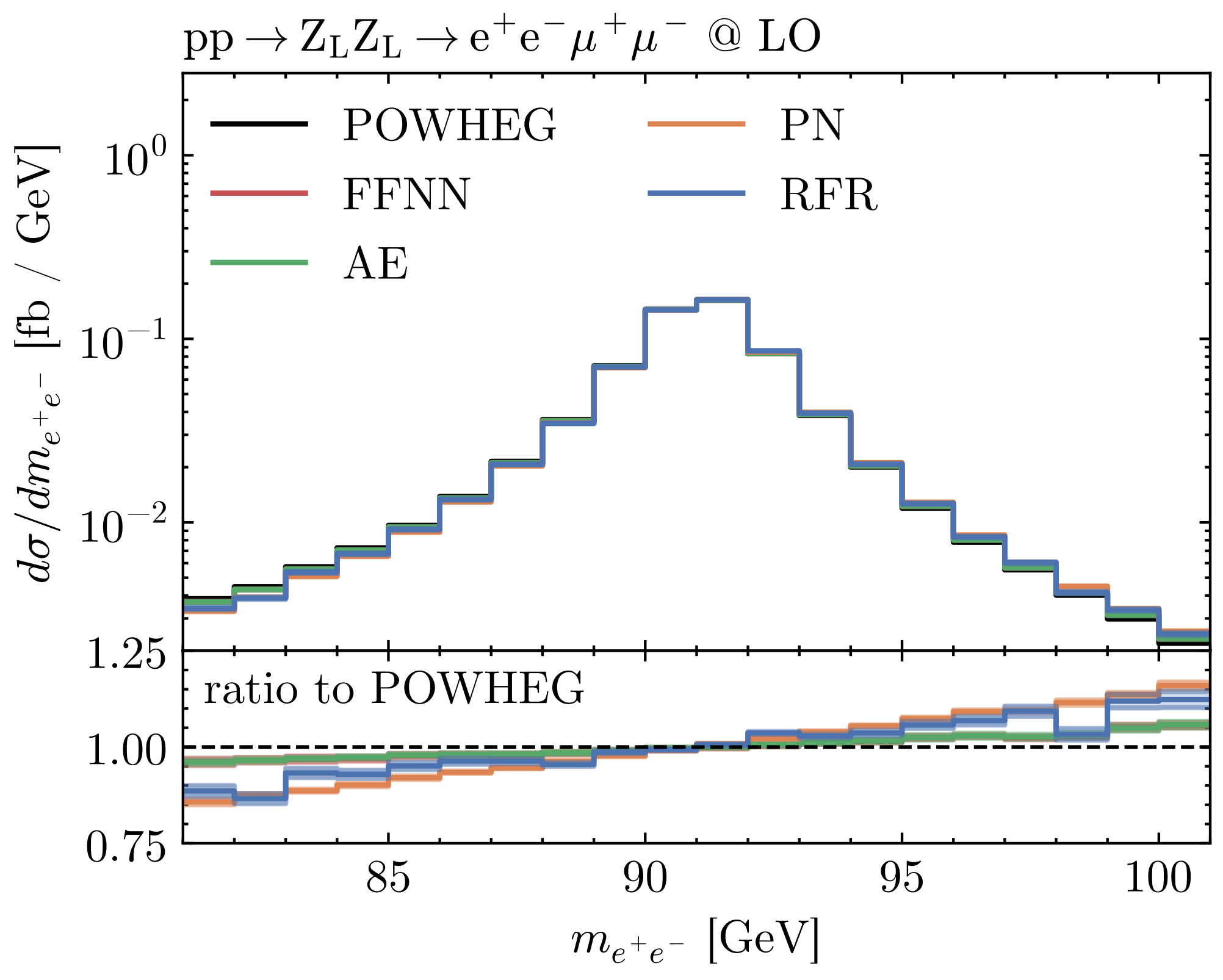
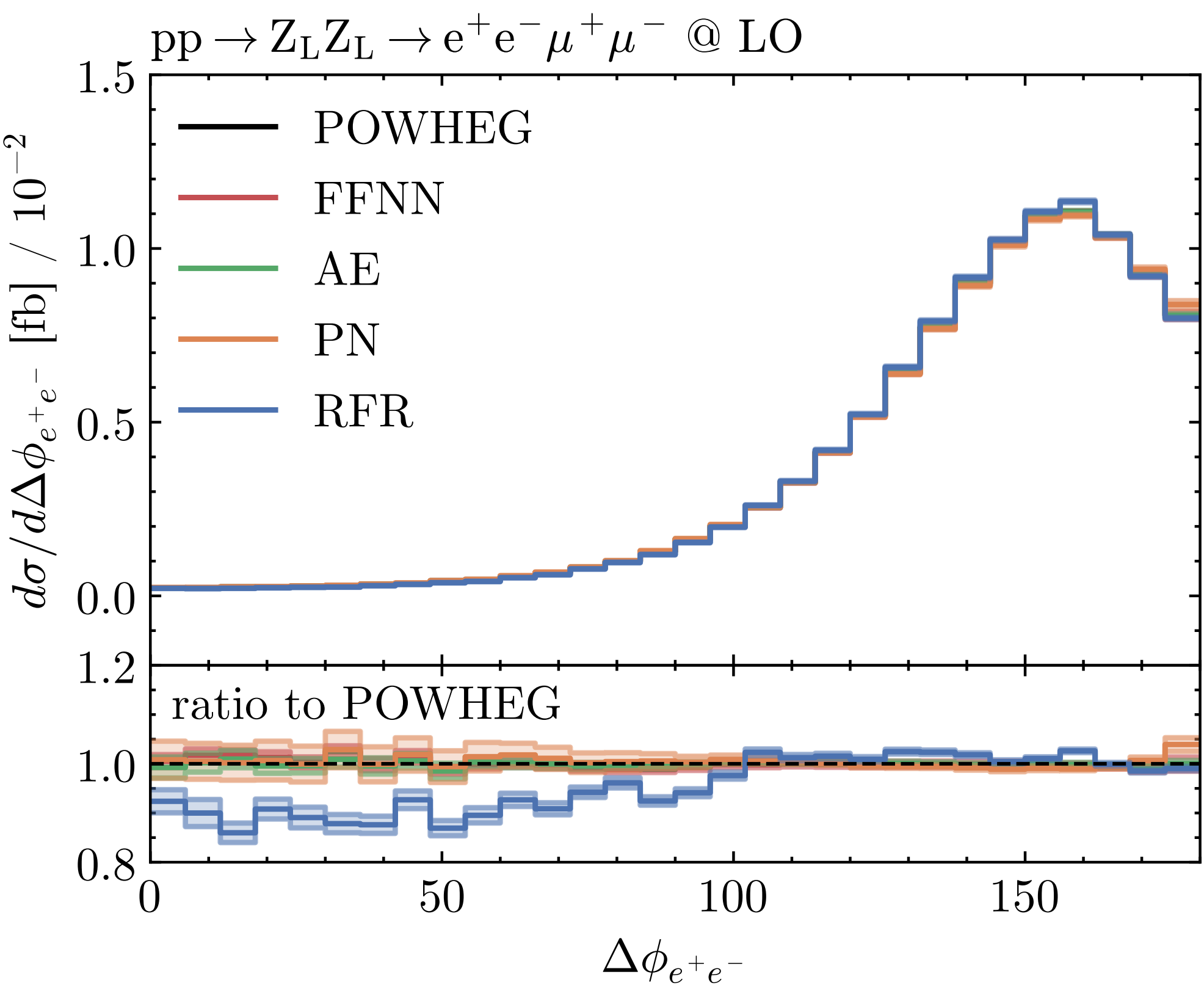
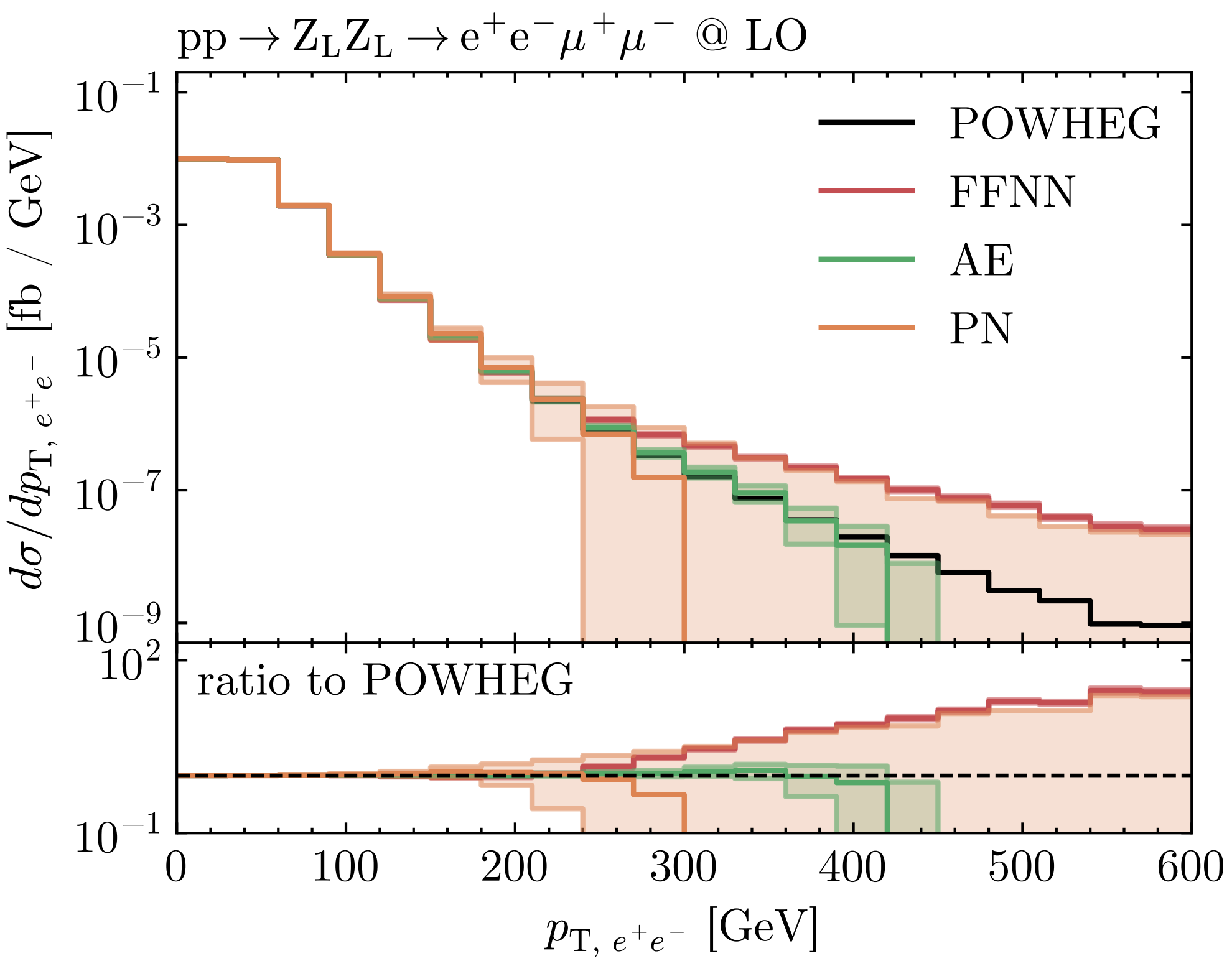

$pp \to Z_L Z_L \to e^+ e^- \mu^+ \mu^-$ @ LO
POWHEG
FFNN
AE
PN
RFR
$d\sigma/d\cos\theta^*_{e^+}$ [fb]
0.0
0.2
0.4
0.6
0.8
ratio to POWHEG
1.0
1.1
−1.0
−0.5
0.0
0.5
1.0
$\cos\theta^*_{e^+}$

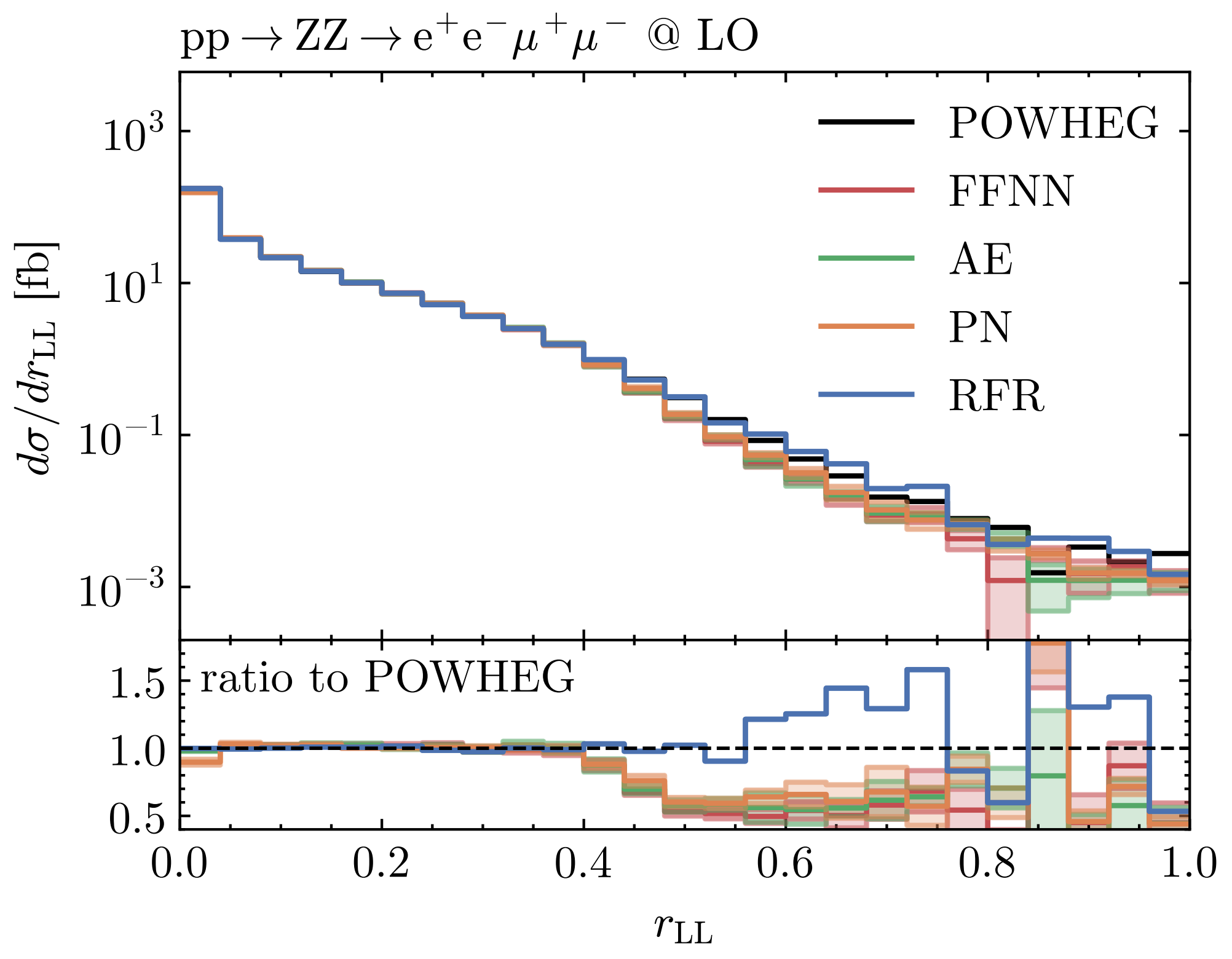

pp → Z_L Z_L → e+ e− μ+ μ− @ LO
POWHEG
FFNN
AE
PN
RFR
dσ/dp_T, e+ [fb / GeV]
ratio to POWHEG
p_T, e+ [GeV]
10^−1
10^−3
10^−5
10^−7
10^−9
10^1
10^−1
0
100
200
300
400

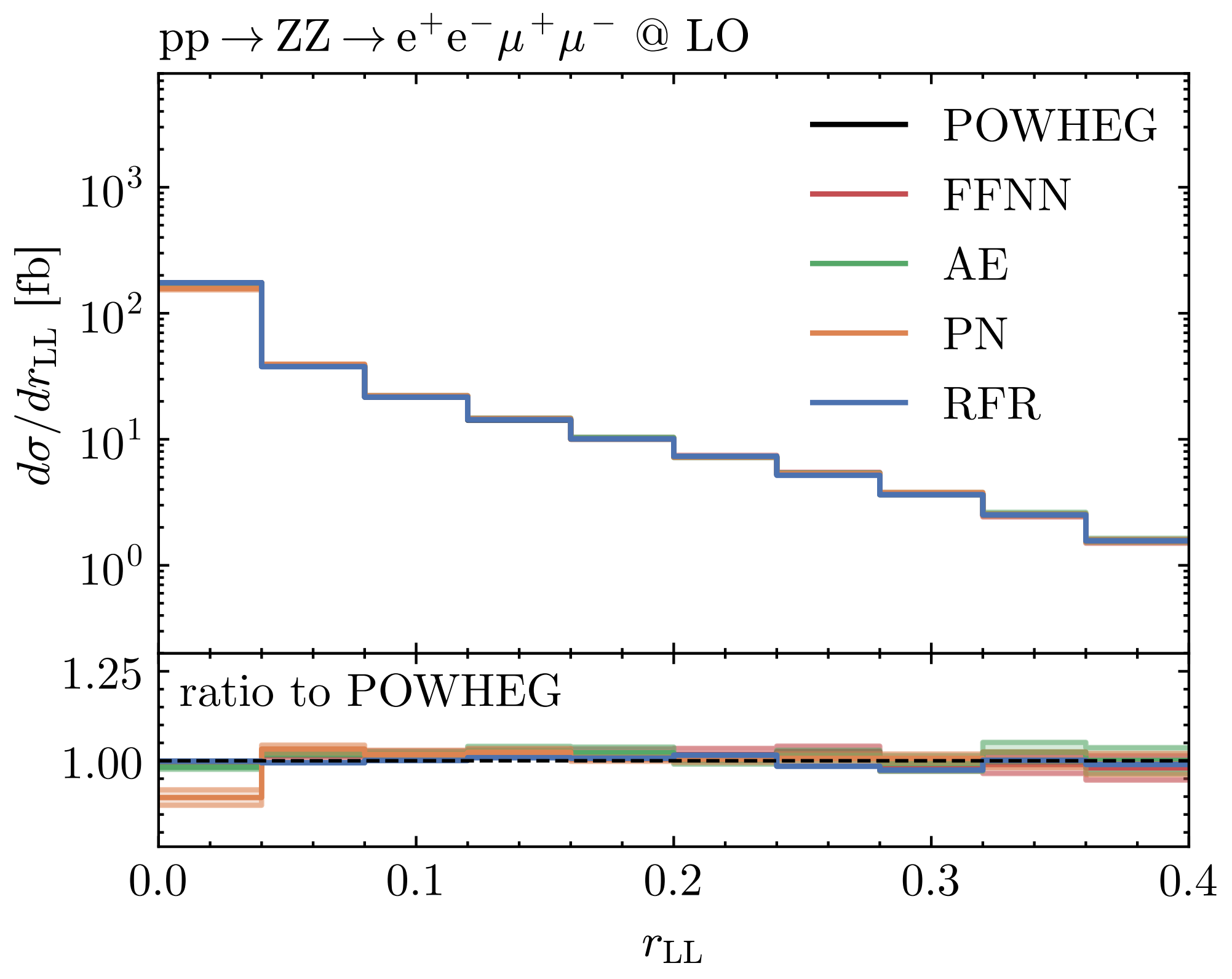

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ LO
POWHEG
FFNN
AE
PN
RFR
$d\sigma/dy_{e^+}$ [fb]
0.3
0.2
0.1
0.0
ratio to POWHEG
1.1
1.0
0.9
−2
−1
0
1
2
y_{e^+}


pp → Z_L Z_L → e+ e− μ+ μ− @ LO
POWHEG
FFNN
AE
PN
RFR
dσ/dm_{e+e−} [fb / GeV]
10^0
10^−1
10^−2
ratio to POWHEG
1.25
1.00
0.75
85
90
95
100
m_{e+e−} [GeV]


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ LO
POWHEG
FFNN
AE
PN
RFR
$d\sigma / d\Delta\phi_{e^+e^-}$ [fb] / 10^{-2}
1.5
1.0
0.5
0.0
1.2
ratio to POWHEG
1.0
0.8
0
50
100
150
$\Delta\phi_{e^+e^-}$


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ LO
POWHEG
FFNN
AE
PN
$d\sigma/dp_{\mathrm{T},\, e^+e^-}$ [fb / GeV]
ratio to POWHEG
$p_{\mathrm{T},\, e^+e^-}$ [GeV]


pp → ZZ → e+e−μ+μ− @ LO
POWHEG
FFNN
AE
PN
RFR
dσ/dr_LL [fb]
10^3
10^1
10^−1
10^−3
ratio to POWHEG
1.5
1.0
0.5
0.0
0.2
0.4
0.6
0.8
1.0
r_LL


pp → ZZ → e+e−μ+μ− @ LO
POWHEG
FFNN
AE
PN
RFR
dσ/dr_LL [fb]
10^3
10^2
10^1
10^0
ratio to POWHEG
1.25
1.00
0.0
0.1
0.2
0.3
0.4
r_LL